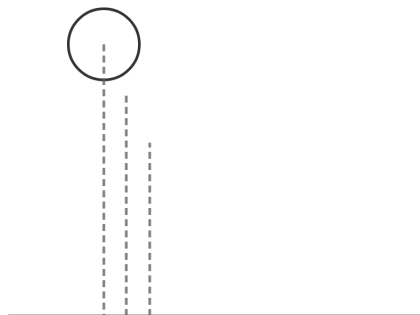


2019A F=ma Exam: Problem 12

Kevin S. Huang



When the ball is launched with velocity v , it reaches its peak in time t from kinematics:

$$\begin{aligned} v - gt &= 0 \\ t &= \frac{v}{g} \end{aligned}$$

By symmetry, the time between bounces is twice this time:

$$T = 2t = \frac{2v}{g}$$

Since the velocity is reduced by a factor of α after each bounce we have

$$T(n) = \frac{2v(n)}{g} = \frac{2\alpha^n v_0}{g}$$

Note that this is the time between successive bounces as a function of **bounce number**. The time from the start to the n th bounce is

$$t(n) = \sum_{k=0}^{n-1} T(k) = \sum_{k=0}^{n-1} \frac{2\alpha^k v_0}{g} = \frac{2v_0}{g} \sum_{k=0}^{n-1} \alpha^k = \frac{2v_0}{g} \frac{1 - \alpha^n}{1 - \alpha}$$

Thus we have

$$(1 - \alpha)t = \frac{2v_0}{g} - \frac{2\alpha^n v_0}{g} = \frac{2v_0}{g} - T$$

$$\boxed{T(t) = \frac{2v_0}{g} - (1 - \alpha)t}$$

so the time between bounces decreases linearly as a function of **time**. The answer is B.